

SQL4Business: Deliverable 2

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Task and model. We preserve Deliverable 1: answer punctual and combined business questions by executing the necessary SQLite queries and producing a numerically faithful executive report. We commit to `Qwen/Qwen2.5-Coder-3B-Instruct` because it is the smallest candidate, matches the best local baseline accuracy (46.7%), and fits faster iteration on a T4. Qwen-7B costs more parameters without a local advantage; Llama-8B is larger, gated, and also scores 46.7% locally.

Intervention. Direct prompting failed on all 5 combined questions. The solution separates planning from execution: a model emits a strict JSON plan; a schema retriever supplies only relevant tables; a read-only SQLite tool executes each step with up to two bounded repairs; deterministic code performs growth, comparison, share, and filter-then-rank operations; a final model report is checked against verified values. The direct baseline and solution use the same 15 questions and SQL-result correctness criterion.

Manager question → relevane schema → JSON plan → read-only SQLite steps → deterministic composition → verified English report

Evaluation set. 15 fixed questions from `data/questions.json`: 10 punctual and 5 combined. The database is synthetic but reproducible, contains sales and inventory from March–August 2026, and is unchanged between baseline and solution. Equivalent SQL is judged by executed rows rather than SQL string equality.

System	Puntual	Combinada	Global
Direct baseline (Qwen-3B)	60.0%	0.0%	40.0%
Structured solution	20.0%	20.0%	20.0%

Alternative and limits. The evaluated alternative is direct prompting with the full schema and no decomposition. The structured solution still depends on the model selecting valid dates, joins, and operation types; a repair budget cannot guarantee recovery. **Failure case: q01:** the solution failed its correctness checks. It is reported from a real trace without replacing its output with gold data.

Reproduction. Run `notebooks/deliverable2_demo.ipynb` on a Colab T4, or execute `scripts/evaluate_baseline_solution.py` from the repository. The README documents dependencies, paths, traces, hardware, and output files.